

SATS: Scenario-Anchored Topology-Inspired Structural Scoring

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Abstract

Sentence-level structural reasoning, reflected in relational connectivity patterns induced by attention distributions, plays an important role in grammatical acceptability judgment and figurative expression understanding. However, most existing scoring approaches rely primarily on token-level probability signals or sentence-level similarity, which do not explicitly capture structural relationships between tokens. We propose Scenario-Anchored Topology-Inspired Structural Scoring (SATS), a structure-aware scoring method that evaluates coherence between a shared scenario and multiple candidate interpretations using connectivity patterns derived from attention matrices. Unlike prior topology-inspired approaches that focus on pairwise comparisons, SATS reformulates structural comparison as scenario-anchored multi-option scoring while maintaining low computational cost. Experiments on benchmark datasets including BLiMP, FLUTE, and Pragmatics show that SATS improves encoder-based model performance in many settings and can approach the accuracy of much larger language models. These results suggest that explicit structural comparison can provide a useful complementary signal for structural language understanding tasks.

1 Introduction

Many language understanding tasks require reasoning over relationships between words rather than relying only on surface lexical similarity (Barzilay and Lapata, 2005). However, standard scoring methods used with recent language models often rely strongly on token-level probability signals (Salazar et al., 2020), which may be insufficient for tasks requiring sentence-level structural coherence.

These limitations are particularly visible in figurative expressions, where meaning arises from interactions among words rather than from literal interpretation. For example, as illustrated in Figure 1,

Mark asked his mom what she thought about his new girlfriend. She replied: "This young lady is 100% polyester."
What does she mean?

1. His girlfriend wore clothes made of polyester.

2. His girlfriend has a beautiful smile.

3. His girlfriend is made of polyester.

4. The girl made a good impression on Mark's mom.

5. His girlfriend's behavior was not very natural.

Figure 1: This example illustrates a metaphor instance from the Pragmatics dataset (Hu et al., 2023). Green denotes the intended interpretation, while pink highlights a commonly favored unintended reading.

the sentence "This young lady is 100% polyester" does not literally describe the material composition of the woman, but instead suggests that her behavior was artificial or unnatural. However, models can misinterpret such expressions when scoring is driven mainly by surface lexical cues such as polyester (McCoy et al., 2019). This pattern suggests that token-level cues alone are often insufficient for capturing relational semantic structure.

Recent research interprets Transformer attention patterns as signals of relational structure between tokens (Clark et al., 2019). From this perspective, attention provides a useful basis for analyzing interaction structures within a sentence. Some studies further examine these structures through graph-based or topological perspectives (Su et al., 2026; Proskurina et al., 2023). However, existing topology-inspired approaches mainly focus on pairwise comparison between two representations, which makes them less suitable for multi-option inference settings where several candidate interpretations must be evaluated under the same shared scenario. In addition, exhaustive exploration over all layer-head pairs can be computationally expensive. To address these limitations, we propose an efficient structure-aware scoring method that

leverages attention-induced connectivity patterns for scenario-anchored multi-option inference.

Our main contributions are as follows. (1) We propose a structure-aware scoring method that evaluates scenario–option relationships using token-level connectivity patterns within attention-induced graphs. (2) We design an algorithm that extends beyond pairwise comparison and applies to multi-option inference problems under a shared scenario. (3) We introduce an efficient computation strategy that avoids exhaustive exploration of the full attention structure, while enabling practical evaluation on grammatical acceptability, figurative interpretation, and pragmatic reasoning tasks.

2 Related Work

2.1 Representation-based Scoring for Language Understanding

Many language understanding tasks require selecting the most plausible interpretation from multiple candidates. A common approach is to score candidate sentences using language model probabilities. For example, Salazar et al. (2020) introduced pseudo–log–likelihood scoring with masked language models for sentence evaluation. Representation-based methods instead compare candidates through semantic similarity between sentence embeddings. Sentence-BERT (Reimers and Gurevych, 2019) enables efficient similarity computation, and subsequent work establishes sentence embeddings as strong tools for semantic comparison and evaluation (Gao et al., 2021; Muenighoff et al., 2023).

Recent work also examines whether Transformer attention encodes structural relationships between tokens. Prior studies show that attention patterns can capture syntactic and coreference relations (Clark et al., 2019) and can be analyzed to interpret token interactions across attention heads (Voita et al., 2019; Vig, 2019). Topological data analysis has further been applied to attention graphs to extract structural features (Kushnareva et al., 2021; Cherniavskii et al., 2022). However, these approaches often require examining many layers and heads, which can be computationally expensive in multi-choice inference settings. In contrast, our work proposes a structure-aware scoring method based on inter-token connectivity that avoids exhaustive comparison over all candidate pairs.

2.2 Figurative Expression Understanding

Figurative expressions remain challenging for language models because they require interpreting meanings beyond literal word semantics (Liu et al., 2022). Most prior work treats figurative language as a detection problem. For example, Li et al. (2023) detect metaphors by modeling discrepancies between literal and contextual meanings, while Yang et al. (2024b) use large language models to identify verb metaphors. Figurative language is also relevant to safety research because metaphorical expressions can implicitly convey harmful content (Zeng et al., 2025; Yan et al., 2025).

Several datasets support research on figurative language. VUA provides token-level annotations of metaphor usage (Leong et al., 2020), while FLUTE introduces a natural language inference benchmark covering metaphor, simile, idioms, and satire (Chakrabarty et al., 2022). Related multiple-choice interpretation settings have also been explored in broader pragmatic language understanding benchmarks (Hu et al., 2023). In contrast to prior work centered on lexical or contextual signals, our approach studies figurative understanding from a structural perspective by evaluating relational coherence between scenarios and candidate interpretations.

3 Methodology

3.1 Background: Representation Topology Divergence

Representation Topological Divergence (RTD), proposed by Barannikov et al. (2022), measures structural differences between two representations across multiple filtration scales. In NLP, Cherniavskii et al. (2022) reinterpret RTD as a graph comparison method over Transformer attention maps.

Given an input sequence of length m , an attention matrix $W \in [0, 1]^{m \times m}$ extracted from a layer-head pair is interpreted as a weighted graph

$$G = (V, E, w), \quad (1)$$

$$V = \{1, \dots, m\}, \quad w : E \rightarrow [0, 1].$$

RTD compares two attention-induced graphs by applying threshold-based filtration and measuring how their connectivity structures differ across filtration levels. This process yields a collection of birth–death intervals, and the RTD score is computed as

$$\text{RTD}(G_a, G_b) = \sum_{\alpha \in \text{Barcode}(G_a, G_b)} (d_\alpha - b_\alpha). \quad (2)$$

A larger RTD value indicates greater structural divergence between the two graphs. Further details on the filtration procedure and the RTD baseline used in our experiments are provided in Appendix A.

Although RTD provides a principled topology-inspired way to compare attention structures, it is primarily formulated as a pairwise graph-comparison method. In multi-option reasoning, several candidate interpretations must be evaluated with respect to the same shared scenario. The goal is therefore not only to compare two graph representations, but to assign each candidate an option-specific score under a common scenario anchor. This motivates SATS, which does not directly extend the RTD score but instead reformulates topology-inspired structural comparison as scenario-anchored multi-option scoring.

3.2 Scenario-Anchored Topology-Inspired Structural Scoring (SATS)

In multi-option reasoning tasks, the goal is to evaluate how well each candidate interpretation structurally aligns with a shared scenario. To this end, we propose Scenario-Anchored Topology-Inspired Structural Scoring (SATS), which anchors structural comparison to the scenario and assigns each option an independent structural score.

Following Carlsson and Vejdemo-Johansson (2022), we interpret attention matrices as weighted graphs and analyze connectivity under varying thresholds. Given a threshold τ , we define the thresholded graph

$$E_\tau = \{\{a, b\} \in E \mid w_{ab} \geq \tau\}, \quad G_\tau = (V, E_\tau). \quad (3)$$

As the threshold decreases, additional edges appear, forming a filtration of graphs. Since we are only interested in connectivity, we consider the 0-dimensional homology H_0 , which depends solely on the 1-skeleton of the graph. Therefore, in practice, the computation can be performed directly on the graph without explicitly constructing the simplicial complex.

For each instance, we concatenate the scenario S with each option O_i to form a token sequence of length T . The attention matrix for layer ℓ and head h is denoted as

$$A_{\ell, h}^{(i)} \in \mathbb{R}_{\geq 0}^{T \times T}. \quad (4)$$

For notational simplicity, the indices (i, ℓ, h) are

omitted hereafter. To eliminate directional asymmetry, we symmetrize the matrix:

$$U = \frac{1}{2}(A + A^\top), \quad \text{diag}(U) = 0. \quad (5)$$

To focus on scenario-option interactions, we treat the set of scenario tokens collectively as a shared anchor $[S]$, and connectivity is analyzed relative to this anchor set. Let o denote an option token. The threshold at which o becomes connected to $[S]$ in the filtration can be expressed as

$$d(o) = \max_{s \in S} \max_{p: s \leadsto o} \min_{e \in p} w_e \quad (6)$$

which corresponds to the widest-path value. Intuitively, this value captures the earliest threshold at which an option becomes structurally connected to the scenario via strong interaction paths.

This formulation serves as an ideal filtration-based connectivity value that motivates SATS, rather than the quantity exhaustively computed in the implementation. Computing the widest-path value for all scenario-option token pairs across all layer-head graphs is expensive. We therefore use a finite Katz index (Katz, 1953) as a practical surrogate that captures cumulative multi-hop structural connectivity through short-to-medium paths. Although it does not exactly recover the death value, it preserves the threshold-based connectivity intuition underlying our formulation and serves as a topology-inspired scoring mechanism in practice.

$$\begin{aligned} \Phi_K(U; \beta) &= \sum_{t=1}^K \beta^{t-1} U^t, \\ U_{\text{blend}} &= \lambda U + (1 - \lambda) \Phi_K(U; \beta), \\ u(o) &= \text{Agg}(\{(U_{\text{blend}})_{s,o}\}_{s \in S}), \quad o \in O_i \\ \text{SATS}(i) &= \frac{1}{|O_i|} \sum_{o \in O_i} -\log(\max(u(o), \epsilon)) \end{aligned} \quad (7)$$

where S and O_i denote the scenario-token index set and the token set of the i -th option, respectively. The terms $\Phi_K(U; \beta)$ and U_{blend} are matrix-valued connectivity terms in $\mathbb{R}_{\geq 0}^{T \times T}$. For each option token o , $u(o)$ is a scalar connectivity value obtained by aggregating the scenario-token connectivity values $\{(U_{\text{blend}})_{s,o}\}_{s \in S}$. We use top- k pooling for Agg by default. The final value $\text{SATS}(i)$ is an option-level scalar score, and the candidate with the lowest score is selected. The final prediction is obtained by

$$\hat{i} = \arg \min_i \text{SATS}(i). \quad (8)$$

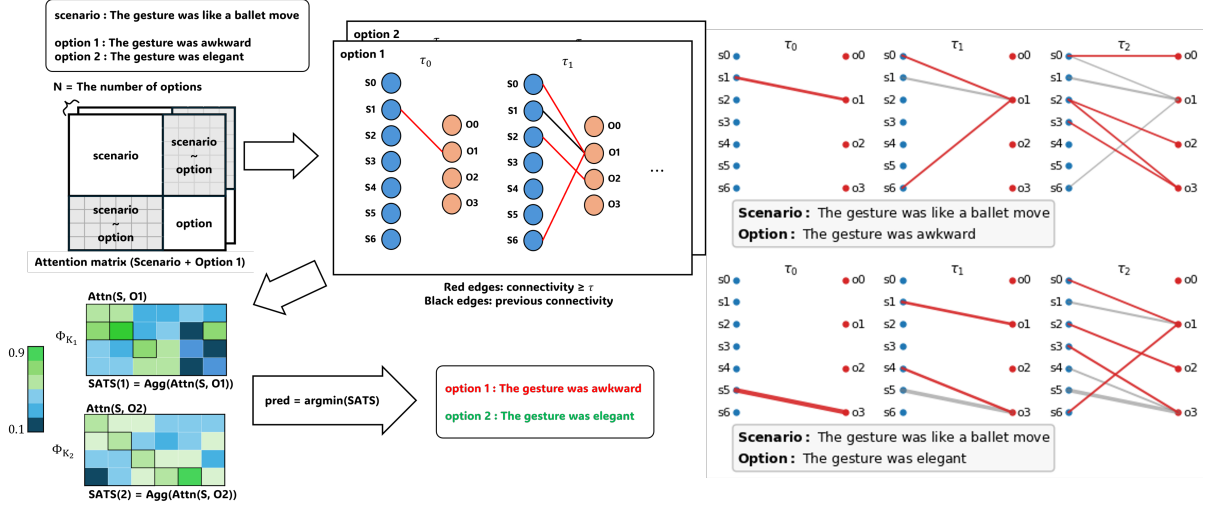


Figure 2: The scenario serves as a shared anchor for all candidate options, and the structural connectivity between the scenario and each option (S–O) is computed to derive the SATS score. The right panel illustrates an example of token connectivity formation obtained using DeBERTa-v3-large, showing how connections between scenario and option tokens emerge as the threshold τ decreases.

The default hyperparameters are $K = 5$, $\beta = 0.5$, $\lambda = 0.5$, and $k = 5$. Hyperparameters are optimized independently with search ranges $K, k \in \{1, \dots, 9\}$ and $\beta, \lambda \in \{0.1, \dots, 0.9\}$. Each step requires K multiplications of the $T \times T$ matrix U , yielding computational complexity $O(KT^3)$. The latency is primarily determined by K and the token length T , while β , λ , and the aggregation choice do not affect the leading-order term.

Illustrative Example. Figure 2 provides an illustrative example of the connection structure formed between scenario and option tokens by SATS with DeBERTa-v3-large on a simile instance. The right panel shows how cross-token edges progressively emerge as the threshold τ decreases. Since $\tau \in [0, 1]$ controls edge activation based on the multi-hop score Φ , higher-weight connections appear earlier at larger threshold values. Earlier connections increase $u(o)$ and therefore lead to lower SATS scores. In this example, Option 2 forms a cross-token connection earlier in the filtration process and ultimately receives the lowest SATS score. The bar graph also shows that, although Option 1 forms an early connection via gesture, its key token awkward receives a relatively low $u(o)$ value. By contrast, Option 2 forms an early connection between s_5 (ballet) and o_3 (elegant). This example illustrates that SATS can favor options whose connectivity patterns align earlier and more broadly with the scenario, rather than depending solely on local lexical overlap in this case.

4 Experiment

4.1 Datasets

We evaluate SATS on three benchmark datasets: BLiMP (Warstadt et al., 2020) for grammatical acceptability, FLUTE (Chakrabarty et al., 2022) for figurative expression understanding, and Pragmatics (Hu et al., 2023) for multi-option reasoning. BLiMP consists of 67 linguistic phenomena grouped into 12 superordinate categories. All experiments are conducted in a zero-shot setting without task-specific training. For SATS, auxiliary data is used exclusively for selecting layer-head pairs and tuning hyperparameters, with no parameter updates. All evaluations are conducted on held-out test data disjoint from the auxiliary data. For the BLiMP dataset, the auxiliary data configuration follows the setup introduced by Cherniavskii et al. (2022). Since the Pragmatics dataset does not provide a training split, we adopt five-fold cross-validation, using 20% of the data as auxiliary data and 80% for evaluation, repeated over three random seeds.

4.2 Models

We evaluate SATS using encoder-based models and open-source LLMs. For encoders, we use RoBERTa-large, RoBERTa-large-MNLI (Liu et al., 2019), and DeBERTa-v3-large (He et al., 2021) as backbones for PLL, RTD, and SATS. For LLMs, we evaluate models from the LLaMA family (Llama-3.2-1B, Llama-3.1-8B, Llama-3.1-70B)

Model	Params	Overall	ANA. AGR	ARG. STR	BINDING	CTRL. RAIS	D-N ARG	ELLIPSIS	FILLER. GAP	IRREGULAR	ISLAND	NPI	QUANTIFIERS	S-V ARG
<i>LLMs</i>														
Qwen3	32B	75.0	90.0	62.9	80.9	71.0	79.1	74.5	69.0	77.4	73.6	75.6	83.3	81.0
Mixtral	141B	78.2	99.4	69.6	78.8	73.6	90.7	84.5	73.9	89.5	67.6	76.9	83.8	81.6
Qwen3	4B	75.7	97.4	65.8	77.3	70.4	86.7	82.6	72.8	74.2	78.6	70.4	78.1	73.6
Llama-3.1	8B	78.4	98.8	72.2	81.0	79.0	90.5	86.2	73.8	92.2	71.0	69.6	75.2	82.0
Qwen2.5	1.5B	78.7	98.0	71.7	76.3	79.0	91.0	83.9	74.1	90.1	73.5	77.1	72.8	81.6
Llama-3.1	70B	78.8	99.4	72.0	83.7	78.1	90.1	84.1	73.0	87.8	71.0	71.6	75.8	85.2
Llama-3.2	1B	79.1	99.4	73.2	78.0	81.1	90.9	88.4	78.2	91.9	68.3	72.5	77.2	82.1
SmolLM2	360M	79.6	98.4	72.2	75.3	82.8	94.5	88.9	76.1	92.8	68.2	72.3	79.6	87.2
Mistral	7B	80.5	99.7	72.1	85.4	78.1	91.1	87.1	75.3	92.8	70.9	80.2	75.0	85.3
<i>Encoder Baselines</i>														
RoBERTa	355M	85.2	97.8	79.1	78.5	82.5	96.2	91.0	88.9	96.8	83.4	83.3	70.2	90.4
RoBERTa-MNLI	356M	47.1	59.4	49.2	33.7	52.3	45.1	36.3	40.1	49.2	41.4	52.1	60.8	45.7
DeBERTa	418M	30.1	11.4	48.5	22.5	35.5	32.0	27.1	28.4	41.8	26.0	27.1	30.0	30.5
<i>RTD</i>														
RoBERTa	355M	82.8	97.4	78.5	85.2	76.5	88.5	80.0	79.7	90.0	76.2	80.5	83.2	92.6
RoBERTa-MNLI	356M	83.1	93.7	75.4	83.4	76.4	96.1	78.5	78.0	98.3	73.7	81.4	83.1	95.4
DeBERTa	418M	82.4	96.9	73.3	80.2	74.0	88.9	85.1	79.8	93.7	80.4	84.1	83.5	90.1
<i>SATS (Ours)</i>														
RoBERTa	355M	94.6	99.3	83.0	93.9	91.1	98.1	93.2	99.1	97.3	95.7	98.8	99.5	94.7
RoBERTa-MNLI	356M	92.9	96.9	79.9	95.5	92.0	90.3	96.0	98.4	98.4	94.1	98.8	99.7	87.9
DeBERTa	418M	95.9	99.8	83.4	97.5	90.1	97.1	98.9	99.8	99.5	97.8	100.0	99.6	94.5

Table 1: Experimental results (accuracy, %) on BLiMP dataset. In the Overall column, **bold** indicates the best result and underlining indicates the second-best result.

(Meta, 2024b,a), the Mistral family (Mistral-7B-v0.3, Mixtral-8x22B-v0.1) (Mistral AI, 2024a,b), the Qwen family (Qwen2.5-1.5B, Qwen3-4B, Qwen3-32B) (Yang et al., 2025, 2024a), and SmolLM2-360M (Allal et al., 2025). These models range from approximately 360M to 141B parameters, and some are evaluated with quantized weights due to memory constraints.

We evaluate all LLMs using base models without instruction tuning for probability-based scoring because instruction tuning is optimized for instruction following and may alter the likelihoods used for candidate scoring. Encoder models are evaluated with pseudo-log-likelihood (PLL) over candidate sentences, whereas causal LLMs are evaluated with conditional log-likelihood (CLL) of each option given the scenario. Because the main LLM evaluation is conducted under a scoring-based protocol rather than a prompt-based generation setting, no task-specific instruction prompt is used. Detailed scoring definitions are provided in Appendix C.

5 Results

5.1 BLiMP

Table 1 reports the results on the BLiMP benchmark, a minimal-pair evaluation of syntactic acceptability (Warstadt et al., 2020). Since the sentence-level scoring protocol strongly affects performance on BLiMP, we evaluate RoBERTa and DeBERTa with PLL and compare them with RTD and SATS

on the same encoder backbones. For consistency, only base versions of open-source LLMs are included in this probability-based scoring evaluation.

Overall, encoder-based models scored with PLL outperform autoregressive LLMs, consistent with prior findings that masked LM-based pseudo-log-likelihood is better suited to grammatical acceptability than left-to-right probabilities (Salazar et al., 2020). RoBERTa-MNLI performs worse than RoBERTa-large, and DeBERTa also shows lower PLL accuracy than RoBERTa, indicating that PLL performance varies substantially across encoder architectures and pretraining objectives.

RTD and SATS provide structural scoring signals derived from attention interactions, rather than relying only on token-level probability scores. Among the evaluated scoring methods, SATS achieves the best overall performance. In particular, DeBERTa with SATS obtains the highest overall accuracy, while RoBERTa with SATS outperforms its PLL-based counterpart. At the category level, DeBERTa with SATS achieves the best results in most categories except Control Raising. This pattern suggests that attention-derived structural connectivity can provide a useful complementary signal for grammatical acceptability judgments.

5.2 FLUTE

Table 2 shows the experimental results on the FLUTE dataset. Since the Sarcasm subtask follows a data structure and evaluation protocol that

Model	Params	Overall	Simile	Metaphor	Idiom
<i>LLMs</i>					
SmolLM2	360M	69.8	65.6	60.5	83.2
Llama-3.2	1B	70.8	64.8	66.1	81.6
Qwen3	4B	72.2	67.2	68.6	80.8
Qwen2.5	1.5B	73.2	72.0	66.1	81.6
Mistral	7B	74.3	76.8	61.3	84.8
Qwen3	32B	74.6	75.2	62.1	86.4
Mixtral	141B	75.6	76.0	62.1	88.8
Llama-3.1	70B	75.9	71.2	71.0	85.6
Llama-3.1	8B	76.4	74.4	66.9	88.0
<i>Encoder Baselines</i>					
RoBERTa	355M	73.5	72.8	76.6	71.2
RoBERTa-MNLI	356M	70.1	68.8	73.4	68.0
DeBERTa	418M	37.2	28.8	48.4	34.4
<i>RTD</i>					
RoBERTa	355M	65.5	63.2	70.2	63.2
RoBERTa-MNLI	356M	78.1	67.2	83.1	84.0
DeBERTa	418M	73.8	65.6	81.5	74.4
<i>SATS (Ours)</i>					
RoBERTa	355M	71.6	63.2	67.7	84.0
RoBERTa-MNLI	356M	90.1	80.0	93.6	96.8
DeBERTa	418M	<u>88.5</u>	78.4	90.3	96.8

Table 2: Experimental results (accuracy, %) on Simile, Metaphor, and Idiom in the FLUTE dataset. In the Overall column, **bold** indicates the best result and underlining indicates the second-best result.

requires a different comparison setting, this study evaluates only the Simile, Metaphor, and Idiom subtasks. All experiments are conducted across three different seeds, and the average accuracy is reported. LLMs are evaluated using conditional log-likelihood scoring, while encoder-based models are evaluated using PLL-based scoring, following the standard scoring protocol for each model family.

Among LLMs, Llama-3.1 achieves the highest overall accuracy. However, the performance pattern differs across subtasks. Most LLMs obtain relatively high accuracy on the Idiom subtask, suggesting that fixed or conventionalized meanings can be captured effectively from large-scale pretraining corpora. In contrast, performance on the Metaphor subtask is generally lower and more variable across models. This suggests that figurative expressions requiring contextual reinterpretation may not be fully captured by likelihood-based scoring alone, even when the underlying model is large.

For encoder models, the RoBERTa family shows relatively stable performance under PLL scoring, whereas DeBERTa exhibits lower accuracy, a pattern also observed in the BLiMP experiment. Applying RTD improves encoder performance, with notable gains for RoBERTa-MNLI and DeBERTa,

indicating that attention-based structural information can complement probability-based scoring. SATS further strengthens this effect by directly comparing scenario–option structural connectivity. In particular, RoBERTa-MNLI with SATS records the highest overall accuracy, while DeBERTa with SATS also shows strong overall performance and achieves the best result on the Idiom subtask. These results suggest that scenario–option structural connectivity provides a useful signal for figurative expression understanding, especially when the correct interpretation depends on relational coherence rather than surface lexical likelihood alone.

5.3 Pragmatics

Table 3 shows the experimental results on the Pragmatics dataset. We use 5-fold cross-validation repeated over three random seeds, yielding 15 runs in total. LLM-based CLL scoring and encoder-based PLL scoring are deterministic, so their accuracy values remain unchanged across runs. In contrast, SATS depends on the seed through layer–head and hyperparameter selection, and its reported accuracy is averaged over all runs. The standard deviations for the SATS-based models are 9.0, 2.5, and 2.7, respectively, following the model order presented in Table 3.

Model	Params	Overall	Deceits	Humour	IndirectSpeech	Irony	Maxims	Metaphor
<i>LLMs</i>								
SmolLM2	360M	34.6	45.0	28.0	50.0	40.0	30.0	15.0
Qwen3	4B	38.5	45.0	28.0	55.0	56.0	20.0	25.0
Qwen2.5	1.5B	40.8	55.0	36.0	55.0	40.0	50.0	10.0
Llama-3.2	1B	44.6	55.0	36.0	70.0	48.0	40.0	20.0
Qwen3	32B	50.0	50.0	48.0	60.0	60.0	40.0	40.0
Llama-3.1	8B	51.5	65.0	52.0	65.0	48.0	50.0	30.0
Mistral	7B	54.6	75.0	64.0	60.0	48.0	55.0	25.0
Llama-3.1	70B	56.2	70.0	80.0	70.0	48.0	50.0	15.0
Mixtral	141B	59.2	70.0	84.0	60.0	56.0	60.0	20.0
<i>Encoder Baselines</i>								
RoBERTa	355M	38.5	45.0	48.0	35.0	40.0	45.0	15.0
RoBERTa-MNLI	356M	22.3	45.0	20.0	20.0	32.0	10.0	5.0
DeBERTa	418M	13.1	20.0	4.0	10.0	20.0	10.0	15.0
<i>SATS (Ours)</i>								
RoBERTa	355M	41.0	46.2	31.7	55.8	41.3	35.8	35.0
RoBERTa-MNLI	356M	64.6	75.0	46.0	76.7	69.0	67.9	52.9
DeBERTa	418M	<u>60.9</u>	55.8	47.3	79.6	63.3	62.9	56.7

Table 3: Experimental results (accuracy, %) in the Pragmatics dataset. In the Overall column, **bold** indicates the best result and underlining indicates the second-best result.

Unlike the previous experiments, this dataset provides contextual descriptions of the target sub-task within each scenario. Consequently, encoder-based PLL scoring, which relies on sentence-level probability comparison, may have limitations in selecting the appropriate option. Consistent with this observation, encoder-based models evaluated with PLL generally show relatively low accuracy, whereas SATS uses structural connectivity between scenario and option tokens and achieves higher performance.

The results show that encoder-based models equipped with SATS perform strongly across several subcategories. In particular, RoBERTa-large-MNLI with SATS obtains the highest overall accuracy, followed by DeBERTa-v3-large with SATS. However, LLMs sometimes outperform SATS on specific categories. For example, in the Humour subcategory, Mixtral achieves a high accuracy of 64%. In contrast, the same model records substantially lower accuracy in the Metaphor subcategory, suggesting that different models exhibit distinct strengths across pragmatic phenomena.

6 Attention Structure and SATS Configuration

To analyze the robustness of the SATS scoring process and the contribution of attention structure, we conduct permutation-based validation and component ablation experiments.

Permutation We first apply an option-order permutation that randomly changes the order of candidate options. Since SATS selects the option with the lowest score using an arg min decision rule, this change does not affect the score assigned to each option or the final prediction. As expected, the results remain unchanged under option-order permutation. We next apply attention permutation by randomly shuffling the attention maps, which disrupts the original token-level connectivity patterns. As shown in Table 4, this leads to substantial performance drops across all models. For example, the overall accuracy of RoBERTa decreases from 71.6 to 56.9, while RoBERTa-MNLI and DeBERTa drop from 90.1 to 77.0 and from 88.5 to 70.6, respectively. These results suggest that SATS depends substantially on structured connectivity patterns in attention.

No multi-hop In the *No multi-hop* setting, multi-hop propagation is removed, and scores are computed using only direct scenario–option connectivity with top- k aggregation. Although performance does not collapse completely, accuracy consistently decreases relative to the default SATS configuration. For example, the overall accuracy of RoBERTa-MNLI drops from 90.1 to 84.0, and RoBERTa decreases from 71.6 to 69.5. The decrease is especially noticeable in the Metaphor category. These results indicate that direct attention alone can preserve limited semantic correspon-

Model	Overall	Simile	Metaphor	Idiom
<i>SATS (Ours)</i>				
RoBERTa	71.6	63.2	67.7	84.0
RoBERTa-MNLI	90.1	80.0	93.6	96.8
DeBERTa	88.5	78.4	90.3	96.8
<i>Attention Permutation</i>				
RoBERTa	56.9	61.6	50.8	58.4
RoBERTa-MNLI	77.0	70.4	76.6	84.0
DeBERTa	70.6	50.4	74.2	87.2
<i>No multi-hop</i>				
RoBERTa	69.5	64.0	62.1	82.4
RoBERTa-MNLI	84.0	70.4	85.5	96.0
DeBERTa	88.0	77.6	90.3	96.0
<i>Katz-only</i>				
RoBERTa	73.0	64.8	70.2	84.0
RoBERTa-MNLI	89.8	80.8	92.7	96.0
DeBERTa	88.8	80.0	89.5	96.8
<i>Max aggregation</i>				
RoBERTa	70.8	63.2	67.7	81.6
RoBERTa-MNLI	87.2	72.8	93.6	95.2
DeBERTa	85.3	76.0	83.9	96.0

Table 4: Ablation results of SATS on FLUTE, assessing attention structure, Katz propagation, direct blending, and link aggregation.

dence, but multi-hop connectivity provides additional benefits by capturing indirect relationships through intermediate tokens. This suggests that propagated connectivity is an important component of SATS.

Katz-only In the *Katz-only* setting, we remove the explicit direct-attention blending term λU and compute scenario–option relations using only the Katz propagation term $\Phi_K(U; \beta)$. Because $\Phi_K(U; \beta)$ includes the first-order term U , this setting should not be interpreted as removing all first-order connectivity. Rather, it evaluates whether the explicit direct-attention blend provides additional benefit beyond the propagated Katz connectivity. As shown in Table 4, this variant yields performance that is comparable to, and in some cases slightly better than, the default SATS configuration. Nevertheless, we adopt the blended design as the default configuration because it explicitly combines local attention links with propagated connectivity and provides a more balanced formulation across settings.

Max aggregation Finally, we replace the default top- k aggregation with maximum-based aggregation, which considers only the single strongest connection for each option token. As shown in Table 4,

this consistently lowers performance. For example, the overall accuracy of RoBERTa decreases from 71.6 to 70.8, while RoBERTa-MNLI and DeBERTa drop from 90.1 to 87.2 and from 88.5 to 85.3, respectively. This result suggests that a single strongest connection is insufficient to capture scenario–option correspondence. In contrast, top- k aggregation considers multiple relevant links, which is particularly important when semantic cues are distributed across several tokens.

7 Conclusion

We propose Scenario–Anchored Topology-Inspired Structural Scoring (SATS), a structure-aware scoring method that evaluates scenario–candidate coherence through attention–derived connectivity patterns. SATS reformulates topology–inspired structural comparison as scenario–anchored multi–option scoring and avoids exhaustive exploration of the full attention space. Across the evaluated structural language understanding benchmarks, SATS improves encoder–based models in many settings and can approach or exceed the performance of much larger language models in several cases. These results suggest that structural connectivity can provide a useful complementary signal beyond token–level likelihood and surface lexical cues.

8 Limitations

SATS is designed as a topology–inspired structural scoring method for evaluating coherence between a shared scenario and candidate interpretations. It is therefore most suitable for tasks where the correct option depends on structural correspondence between the scenario and candidate options. Tasks that primarily require factual recall, external knowledge, or symbolic problem solving may benefit less from SATS alone, since these settings often require information beyond attention–derived structural connectivity.

The current implementation relies on a practical Katz–based graph surrogate rather than exact persistent homology computation, and it requires auxiliary data for layer–head and hyperparameter selection. In addition, our evaluation is limited to English benchmarks and relatively short sentence–level or scenario–option inputs. The current SATS evaluation is also limited to encoder–based models with bidirectional attention, and its extension to decoder–only architectures remains an open direction.

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A Detailed Formulation of the RTD Baseline

This section provides the detailed formulation of Representation Topological Divergence (RTD) used as a topology-inspired baseline in our experiments. The main text summarizes RTD only to provide the necessary background for SATS, whereas this appendix describes the filtration procedure and the candidate-selection protocol used for the RTD baseline.

Given an input sequence of length m , an attention matrix $W \in [0, 1]^{m \times m}$ extracted from a layer-head pair (ℓ, h) is interpreted as a weighted graph

$$\begin{aligned} G &= (V, E, w), \\ V &= \{1, \dots, m\}, \quad w : E \rightarrow [0, 1]. \end{aligned} \quad (9)$$

RTD applies a sequence of thresholds τ to construct a filtration

$$G_{\tau_1} \supseteq G_{\tau_2} \supseteq \dots \supseteq G_{\tau_T}, \quad (10)$$

where edges are gradually removed as the threshold increases. This filtration tracks how the connectivity structure of the graph changes across threshold levels.

Let G_a and G_b denote two graphs obtained from different input sequences. During filtration, edges appearing in only one graph are treated as births, while shared edges correspond to deaths. This yields a birth–death interval

$$[b_\alpha, d_\alpha] \quad (11)$$

for each edge α , and the collection of these intervals forms a persistence barcode. Since the comparison depends only on connectivity, RTD considers 0-dimensional homology. The length of each bar is defined as

$$\text{length}(\alpha) = d_\alpha - b_\alpha, \quad (12)$$

and the RTD score is computed as

$$\text{RTD}(G_a, G_b) = \sum_{\alpha \in \text{Barcode}(G_a, G_b)} (d_\alpha - b_\alpha). \quad (13)$$

A larger RTD value indicates greater structural divergence between the two attention-induced graphs.

For the RTD baseline used in our experiments, each candidate hypothesis or option H is concatenated with the scenario or source sentence S :

$$X = \text{concat}(S, H). \quad (14)$$

For a binary candidate setting with one semantically implicative candidate and one contradictory candidate, we denote the corresponding graphs by G_E and G_C , respectively. Since RTD is not necessarily commutative, both directional scores are computed:

$$\text{RTD}(G_E, G_C), \quad \text{RTD}(G_C, G_E). \quad (15)$$

Following prior work, the interpretation associated with the smaller directional score is selected:

$$\text{RTD}(G_E, G_C) < \text{RTD}(G_C, G_E). \quad (16)$$

RTD scores are then aggregated across selected attention heads using a head-ensemble strategy.

This baseline provides a principled topology-inspired comparison of attention structures, but it remains a pairwise graph-comparison method. SATS differs by assigning each candidate option an independent structural score relative to the same scenario anchor.

B Connectivity Interpretation and Stability Analysis

This section explains why the practical SATS score can be interpreted as a meaningful approximation of filtration-based structural connectivity. First, we show that the ideal death value in Eq. 6 corresponds to widest-path connectivity between the scenario anchor and an option token. This interpretation indicates that SATS is motivated by bottleneck connectivity, rather than by isolated local attention strength. Second, we analyze the finite Katz-based surrogate used in Eq. 7 and show that the resulting score is Lipschitz continuous with respect to perturbations of the attention matrix. This stability property suggests that small changes in attention weights do not lead to arbitrary changes in the SATS score. Together, these results clarify how the topology-inspired connectivity view supports the practical scoring behavior of SATS.

B.1 Widest-Path Interpretation of the Death Value

Recall that the death value defined in Eq. 6 is

$$d(o) = \max_{s \in S} \max_{p: s \rightsquigarrow o} \min_{e \in p} w_e.$$

This quantity corresponds exactly to the classical widest-path connectivity between the scenario node and the option token.

Proposition 1. *The death value $d(o)$ equals the widest-path value $\text{wp}([S], o)$.*

Proof. Let

$$\text{wp}(u, v) = \max_{p: u \leadsto v} \min_{e \in p} w_e \quad (17)$$

denote the widest-path value in a weighted graph.

If $\tau > \text{wp}([S], o)$, then every path from $s \in S$ to o contains an edge with weight smaller than τ . Hence o is disconnected from $[S]$ in G_τ . Conversely, if $\tau < \text{wp}([S], o)$, there exists a path p^* such that $\min_{e \in p^*} w_e \geq \tau$.

Thus o is connected to $[S]$ in G_τ . Therefore the threshold at which connectivity first appears coincides with the widest-path value. $\square$

B.2 Stability of the Katz-Based Approximation

Recall that Eq. 7 defines the truncated Katz operator

$$\Phi_K(U; \beta) = \sum_{t=1}^K \beta^{t-1} U^t.$$

We analyze the stability of this operator under perturbations of the attention matrix.

Proposition 2. *The functions $u(o)$ and $\text{SATS}(i)$ are Lipschitz continuous with respect to the matrix U .*

Proof. Consider two symmetric matrices U and V . Using the telescoping identity

$$U^t - V^t = \sum_{j=0}^{t-1} U^{t-1-j} (U - V) V^j \quad (18)$$

we obtain

$$\begin{aligned} \Phi_K(U; \beta) - \Phi_K(V; \beta) &= \sum_{t=1}^K \beta^{t-1} (U^t - V^t) \\ &= \sum_{t=1}^K \beta^{t-1} \sum_{j=0}^{t-1} U^{t-1-j} (U - V) V^j. \end{aligned} \quad (19)$$

Taking norms yields

$$\begin{aligned} &\|\Phi_K(U; \beta) - \Phi_K(V; \beta)\| \\ &\leq \left(\sum_{t=1}^K t \beta^{t-1} M^{t-1} \right) \|U - V\| \end{aligned} \quad (20)$$

where $M = \max\{\|U\|, \|V\|\}$. Therefore the resulting SATS scores are Lipschitz continuous. $\square$

C PLL and CLL Scoring Definitions

For the probability-based baseline, each candidate option is combined with the scenario to form a single input sentence. The resulting input is constructed as $\{\text{scenario}\} \{\text{sep}\} \{\text{option}\}$, where sep denotes a separator token between the scenario and the option. Each candidate sentence is scored independently, and the option corresponding to the highest score is selected as the final prediction.

For encoder-based models, the sentence score is estimated using pseudo-log-likelihood (PLL) for the input sentence $x = (x_1, \dots, x_T)$. PLL is computed by masking each token, evaluating the probability of the original token at the masked position, and averaging these values:

$$\text{PLL}(x) = \frac{1}{T} \sum_{t=1}^T \log P(x_t \mid x_{\setminus t}) \quad (21)$$

where $x_{\setminus t}$ denotes the remaining tokens after masking x_t .

For causal LLMs, the sentence score is computed using the conditional log-likelihood (CLL) of the option given the scenario. Let s denote the scenario and o the option. For the sequence $x = (s, o)$, the CLL is defined as

$$\text{CLL}(o \mid s) = \frac{1}{|o|} \sum_{t=1}^{|o|} \log P(o_t \mid s, o_{<t}) \quad (22)$$

where o_t denotes the t -th token of the option and $o_{<t}$ represents the preceding option tokens. The LLM therefore evaluates each option by sequentially computing the generation probability of option tokens conditioned on the scenario and previously generated option tokens. This scoring protocol should be distinguished from prompt-based LLM judgment. The model is not asked to generate an answer label, and no task-specific instruction prompt is used. Instead, each candidate option is assigned a conditional likelihood score, and the option with the highest score is selected.

D Supplementary Prompt-Based LLM Comparison

The main experiments in this paper compare SATS with scoring-based methods such as PLL, CLL, and RTD. Since SATS assigns an explicit structural score to each candidate option, prompt-based LLM judgment is not used as the primary baseline. Nevertheless, prompt-based LLM judgment can

BLiMP
Question:
Which sentence is grammatical?
Options:
A. {sentence_a}
B. {sentence_b}
Choose the grammatical sentence. Answer with only the option letter.
Answer:
FLUTE / Pragmatics
Question:
{scenario}
Options:
A. {option_a}
B. {option_b}
C. {option_c}
...
Choose the best answer. Answer with only the option letter.
Answer:

Figure 3: Prompt templates used for the supplementary prompt-based LLM comparison.

Model	Params	BLiMP	FLUTE	Pragmatics
SmolLM2	360M	41.6	8.8	8.5
Llama-3.2	1B	49.9	54.5	31.5
Qwen2.5	1.5B	52.4	70.1	49.2
Mistral	7B	55.1	69.8	44.6
Llama-3.1	8B	62.0	84.5	48.5
Qwen3	4B	73.9	88.8	66.9
Mixtral	141B	89.0	82.1	70.0
Llama-3.1	70B	77.3	92.3	76.2
Qwen3	32B	82.2	91.7	74.6

Table 5: Supplementary prompt-based LLM comparison.

provide a useful practical reference point. We therefore conduct a supplementary prompt-based evaluation on the same benchmark datasets. The prompt-based comparison uses the same base LLMs described in Section 4.2.

The prompt templates are shown in Figure 3. For BLiMP, the model selects the grammatical sentence from a minimal pair. For FLUTE and Pragmatics, the model is given a scenario and candidate options and is asked to output the most appropriate option letter. For generation, we use greedy decoding without sampling, restrict the output space to valid option labels (A/B/C/...), and generate only a single label token under constrained decoding. This setup evaluates generation-based option selection and is distinct from the CLL-based LLM scoring used in the main experiments.

Table 5 reports the overall accuracy of prompt-

based LLM judgment on BLiMP, FLUTE, and Pragmatics. The results show that prompt-based generation can benefit substantially from model scale. Large LLMs achieve strong performance under constrained generation, indicating that sufficiently large models can effectively select candidate options from concise prompts. However, this setting differs from the scoring-based protocol studied in the main experiments and typically requires substantially larger models and greater computational resources.

Among prompt-based models below the 10B scale, performance is not consistently stronger than SATS-based encoder scoring. For example, although Qwen3-4B performs strongly on Pragmatics, SATS-based encoder models achieve higher accuracy on BLiMP and FLUTE while using substantially fewer parameters. These results should therefore be interpreted as supplementary context rather than as the primary comparison. The goal of SATS is not to replace large prompt-based LLM judgment, but to provide an efficient structural scoring module that complements likelihood-based scoring when explicit candidate scoring is required or when smaller encoder models are preferred for efficiency.

E Dataset Composition

Table 6 reports the number of auxiliary and test instances used in each dataset and subcategory. As described in Section 4.1, auxiliary data is used only for layer-head selection and hyperparameter tuning, and does not contribute to the reported test accuracy. For BLiMP, all 12 subcategories use identical split sizes, so a single representative entry is shown.

F Auxiliary Data Ratio Sensitivity Analysis

SATS does not update model parameters; auxiliary data is used only for layer-head selection and hyperparameter tuning. While the main experiments report the auxiliary/test split sizes, they do not examine whether SATS is sensitive to the amount of auxiliary data. To address this point, we conduct an auxiliary data ratio sensitivity analysis on the FLUTE dataset. Specifically, we randomly sample 25%, 50%, and 75% of the original auxiliary set, perform layer-head selection and hyperparameter tuning using only each sampled subset, and evaluate the selected configuration on the same held-out test set.

	BLiMP	FLUTE			Pragmatics					
	each subcategory	simile	metaphor	idiom	Deceits	Humour	IndirectSpeech	Irony	Maxims	Metaphor
Auxiliary	100	625	625	885	4	5	4	5	4	4
Test	1000	125	124	125	16	20	16	20	16	16

Table 6: Auxiliary and test split sizes for each dataset and subcategory. For BLiMP, all 12 subcategories use identical split sizes, so a single representative entry is shown.

Model	Overall	Simile	Metaphor	Idiom
<i>SATS</i>				
RoBERTa	71.6	63.2	67.7	84.0
RoBERTa-MNLI	90.1	80.0	93.6	96.8
DeBERTa	88.5	78.4	90.3	96.8
<i>SATS (25%)</i>				
RoBERTa	69.0	60.8	67.7	78.4
RoBERTa-MNLI	90.6	80.8	93.5	97.6
DeBERTa	85.8	77.6	83.9	96.0
<i>SATS (50%)</i>				
RoBERTa	70.8	61.6	66.9	84.0
RoBERTa-MNLI	89.8	80.8	91.9	96.8
DeBERTa	89.3	80.0	91.1	96.8
<i>SATS (75%)</i>				
RoBERTa	72.7	64.8	69.4	84.0
RoBERTa-MNLI	90.4	80.8	93.5	96.8
DeBERTa	89.0	79.2	91.1	96.8

Table 7: SATS performance on FLUTE under different auxiliary-data ratios.

Table 7 reports the results. Reducing the auxiliary ratio does not lead to substantial performance degradation. In some cases, reduced-ratio settings slightly outperform the configuration selected using the full auxiliary set, but this should be interpreted as split-dependent variation rather than evidence that smaller auxiliary sets are systematically preferable. Overall, SATS remains stable across auxiliary-data ratios, suggesting that it does not rely excessively on a specific auxiliary split and that a moderate amount of auxiliary data is sufficient for stable selection and tuning. Thus, the results should be viewed as a sensitivity analysis under reduced auxiliary data, not as an attempt to identify an optimal auxiliary ratio.

G Latency Comparison

Table 8 compares the average latency of each method on the FLUTE and Pragmatics datasets. Each value represents the average time, measured in seconds, required to compute the score for a single sample.

For LLM-based methods, latency generally increases with model size. In particular, Mixtral ex-

	Params	FLUTE	Pragmatics
<i>LLMs</i>			
SmolLM2	360M	0.1621	0.357
Llama-3.2	1B	0.1249	0.220
Qwen2.5	1.5B	0.1525	0.378
Qwen3	4B	0.2046	0.482
Mistral	7B	0.2166	0.428
Llama-3.1	8B	0.2596	0.458
Qwen3	32B	0.9989	2.297
Llama-3.1	70B	2.1175	4.754
Mixtral	141B	6.3351	15.222
<i>Encoder Baselines</i>			
RoBERTa	355M	0.5791	10.091
RoBERTa-MNLI	356M	0.6169	4.769
DeBERTa	418M	1.4914	3.484
<i>RTD</i>			
RoBERTa	355M	0.0750	N/A
RoBERTa-MNLI	356M	0.1423	N/A
DeBERTa	418M	0.0774	N/A
<i>SATS (Ours)</i>			
RoBERTa	355M	0.0866	0.307
RoBERTa-MNLI	356M	0.0954	0.221
DeBERTa	418M	0.0957	0.380

Table 8: Average scoring latency per sample (seconds) on the FLUTE and Pragmatics datasets across different model groups.

hibits substantially higher latency than the other LLMs. Encoder-based PLL methods also incur relatively high latency because they require repeated probability computations over candidate sentences. Latency is also higher on Pragmatics than on FLUTE, since Pragmatics contains more answer options and therefore requires additional scoring computations for each sample.

In contrast, RTD and SATS show lower latency than medium- and large-scale LLM-based methods. Once the hyperparameters are fixed, SATS computes scores directly from extracted attention matrices without repeated sentence-level probability evaluation, which reduces overall computation. As a result, SATS maintains competitive latency, particularly relative to larger LLM baselines. Even on Pragmatics, where latency increases with the number of answer options, SATS remains substantially more efficient than most LLM-based methods.

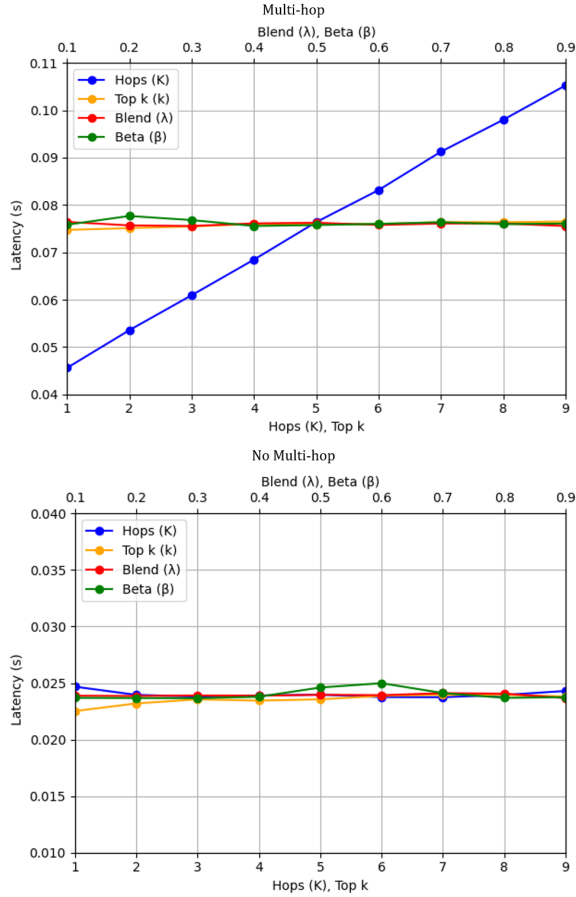


Figure 4: Latency variation on the FLUTE dataset when varying one hyperparameter at a time using DeBERTa. Top: multi-hop configuration. Bottom: non-multi-hop configuration.

Figure 4 analyzes how latency changes when one hyperparameter is varied while the others are held constant. For each parameter, K and top- k are varied from 1 to 9, while λ and β are varied from 0.1 to 0.9. The remaining parameters are fixed to $K = 5$, $k = 5$, $\lambda = 0.5$, and $\beta = 0.5$. The top plot corresponds to the multi-hop configuration, whereas the bottom plot shows the configuration without multi-hop. Both experiments are conducted on FLUTE using DeBERTa.

When multi-hop is disabled, computation mainly consists of aggregation over the attention matrix, and latency remains concentrated around 0.025 seconds. When multi-hop is enabled, the additional iterative hop operations increase latency to approximately 0.075 seconds. Among the hyperparameters, the hop parameter K has the most noticeable effect on latency. Latency increases almost linearly as K grows, since the number of multi-hop propagation steps directly determines the computational workload.

This trend is consistent with the theoretical complexity analysis of SATS. The multi-hop computation in SATS has time complexity $O(KT^3)$, where K denotes the number of hops and T denotes the number of tokens. The observed linear increase in latency with respect to K is therefore consistent with this complexity analysis.

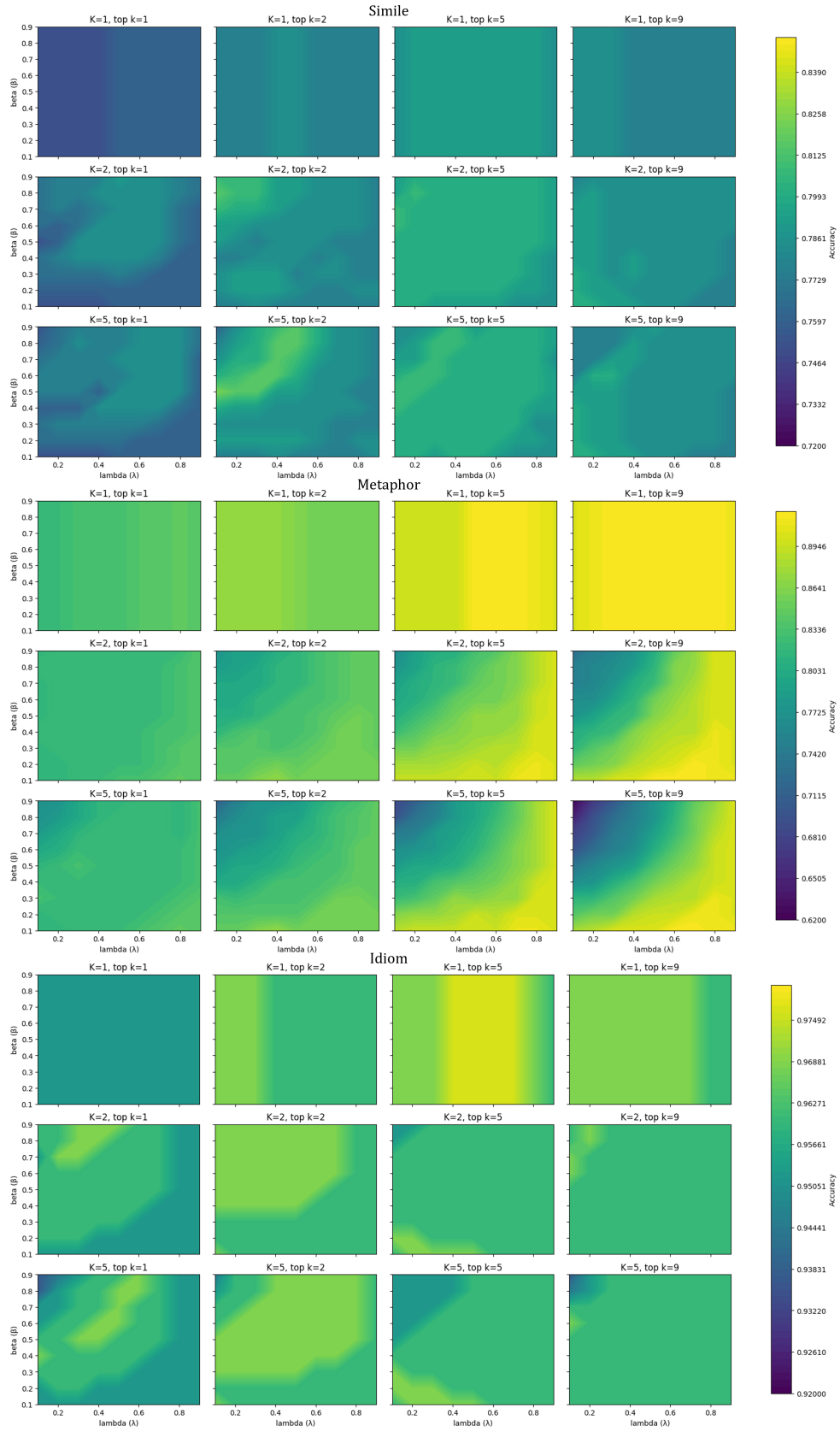


Figure 5: Accuracy across hyperparameter configurations (K , $\text{top } k$, λ , β) on the three FLUTE datasets (Simile, Metaphor, Idiom) using DeBERTa.

H Hyperparameter Sensitivity Analysis

To examine whether the performance of SATS depends heavily on specific hyperparameter settings, we conduct systematic experiments over combinations of K , top- k , λ , and β . Figure 5 shows the accuracy obtained under each hyperparameter configuration across the three FLUTE datasets (Simile, Metaphor, and Idiom) using the DeBERTa model. We set K to 1, 2, and 5, top- k to 1, 2, 5, and 9, and vary λ and β from 0.1 to 0.9.

Overall, SATS maintains relatively stable performance across a wide range of hyperparameter settings. On the Simile dataset, performance variations across configurations are relatively small. A higher accuracy region appears around $K = 5$ and top- $k = 2$, but this region is not confined to a narrow parameter range. This pattern suggests that incorporating a moderate number of hops may help capture the relationships required for simile interpretation.

On the Metaphor dataset, accuracy generally increases as λ grows. However, accuracy drops when β becomes excessively large, even under the same λ . This tendency becomes more pronounced as K increases, suggesting that referencing too many hops increases the likelihood of including semantically less relevant tokens. As a result, selecting and scoring semantically and structurally related tokens appears more beneficial than simply increasing the number of hops.

The Idiom dataset maintains relatively high accuracy overall. Since idioms typically appear as phrases or clauses composed of multiple tokens, structural relations may not be fully captured when top- k is small. This behavior is particularly visible when top- $k = 1$. While certain regions show improved accuracy as K increases by incorporating connections between intermediate tokens, performance may also decline when both K and top- k grow simultaneously, as too many intermediate tokens are referenced.

Overall, these results suggest that the optimal hyperparameter ranges vary depending on dataset characteristics. In addition, a clear interaction between K and top- k is observed: accuracy tends to decline when both parameters increase simultaneously. This pattern likely reflects the interaction among scenario tokens, option tokens, and intermediate tokens within the connection structure.

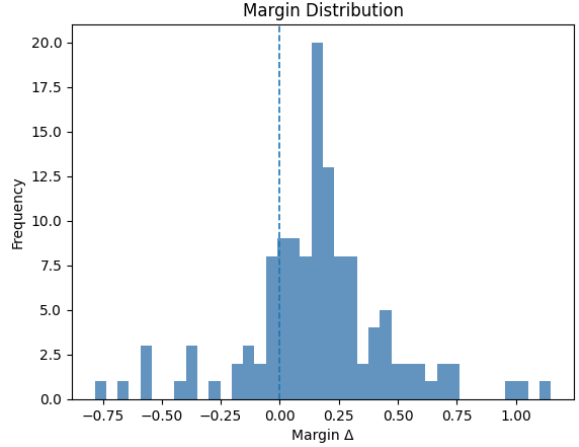


Figure 6: Distribution of SATS margins, defined as the difference between the minimum score of incorrect options and the score of the gold option.

I Margin Distribution of SATS Scores

SATS determines the appropriate response by selecting the option with the lowest score. To quantitatively analyze the score difference between the selected option and the remaining options, we conduct a margin experiment. Figure 6 presents a histogram of the margin distribution computed on the Simile dataset using the DeBERTa-v3-large model. The margin is defined to be positive when the SATS score of the correct option is lower than those of the incorrect options. As shown in the figure, most margin values lie in the positive region, with only a small number of extreme outliers. This indicates that SATS scores serve as a stable relative comparison metric within each problem.

J Qualitative Analysis

J.1 Successful Case

Figure 7 presents an example from the Pragmatics Irony dataset. The instance is a four-way multiple-choice question, with options listed in numerical order. The correct answer is option 4, which attains the lowest SATS score and is therefore correctly selected. To illustrate the scoring behavior, we compare the SATS process of option 4 with that of option 1, which yields the second-lowest score. At the first threshold τ , option 4 exhibits a larger value than option 1, indicating that option 4 becomes structurally connected to the scenario at an earlier stage. This implies that option 4 captures tokens for which $d(o)$ is defined earlier, and these tokens carry sufficient semantic content.

The key semantic cues in the scenario originate

from s_{23} (Class), s_{24} (y), and s_{25} (outfit). In option 1, however, the person names s_{10} (Barb) and s_{30} (Tony) are aligned with the pronoun o_0 (He), while s_{17} (dressed) is aligned with o_5 (dress). This reflects a preference for superficial lexical similarity, which fails to capture the intended structural meaning. In contrast, option 4 establishes connections between s_{23} (Class) and s_{25} (outfit) with o_6 (not) and o_7 (appropriate) at the first threshold τ . These connections directly reflect the ironic interpretation of the scenario. This example demonstrates that SATS captures structurally salient tokens within the scenario and prioritizes options whose internal structure aligns with the intended interpretation.

J.2 Error Case

Figure 8 shows a failure case in which structural connectivity alone is insufficient. SATS assigns a lower score to the incorrect option, option 2, because the option forms strong connectivity through semantically related affective tokens despite containing the phrase o_2 – o_3 (lost all), which reverses the intended meaning. In contrast, the correct option, option 1, preserves the meaning of s_3 (blanketed) through o_3 (covered). The incorrect option also contains semantically related affective tokens, such as o_5 (sadness) and o_7 (sorrow), which form strong scenario–option connectivity. However, lost all expresses the removal of sadness, producing a semantic direction opposite to the scenario. SATS therefore captures the connectivity among related affective tokens but does not explicitly model this polarity reversal. This case suggests that SATS may need to be combined with entailment-aware or contradiction-sensitive signals when candidate distinctions depend on negation, removal, or fine-grained semantic directionality.

K Future Work

Future work can explore tighter connections between the current Katz-based graph surrogate and exact filtration-based topological formulations. Although SATS preserves the connectivity intuition of the filtration view, it does not compute full persistence barcodes or exact death values for all token pairs. Developing more faithful yet efficient approximations may further strengthen the theoretical grounding of topology-inspired structural scoring.

Another important direction is broader evalua-

tion. The current experiments focus on English benchmarks with relatively short sentence-level or scenario–option inputs. Future studies should examine longer-context reasoning, cross-document settings, and typologically diverse languages. In particular, Asian-language settings may provide useful test cases because structural and pragmatic cues can differ substantially from English.

Finally, extending SATS to decoder-only LLMs remains an open direction. The current method relies on bidirectional attention matrices from encoder-based models, whereas decoder-only LLMs use causal attention. Possible extensions include bidirectional attention patterns, hidden-state similarity graphs, or using SATS as a structural reranking module over LLM outputs. Extending SATS with additional semantic or pragmatic signals may also improve its applicability to phenomena such as humor, where structural connectivity alone may be insufficient.

L LLM Usage

We used large language models only as assistive tools for (i) polishing and stylistic refinement of the manuscript, and (ii) surfacing candidate papers during literature search. All substantive content (problem formulation, methods, experiments, analyses, and conclusions) was written by the authors. Any text or citations suggested by an LLM were reviewed and verified by the authors, and we cite only sources we personally inspected.

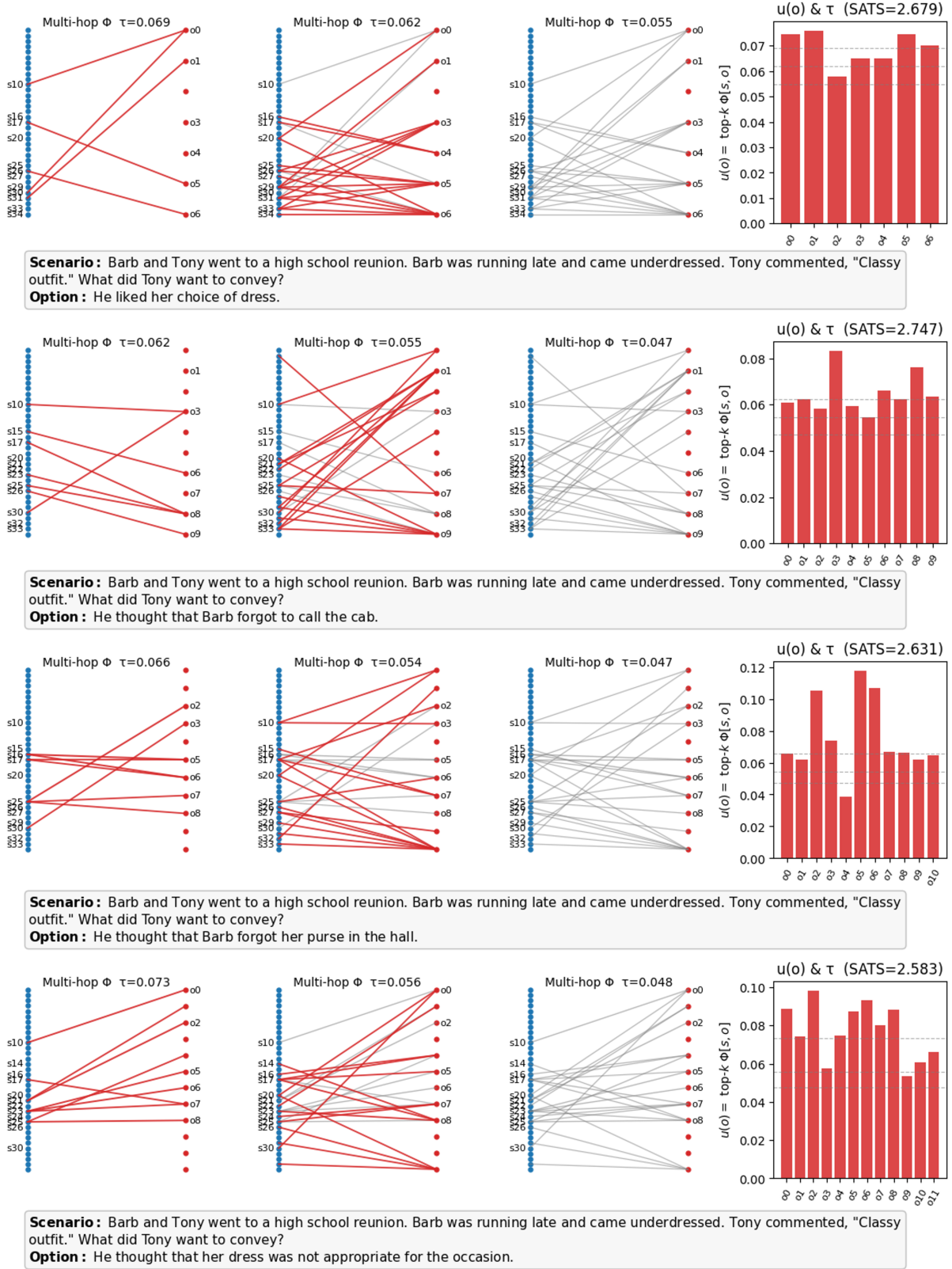


Figure 7: An example of SATS computation for an irony instance, illustrating multi-hop scenario–option interactions and option-level scores.

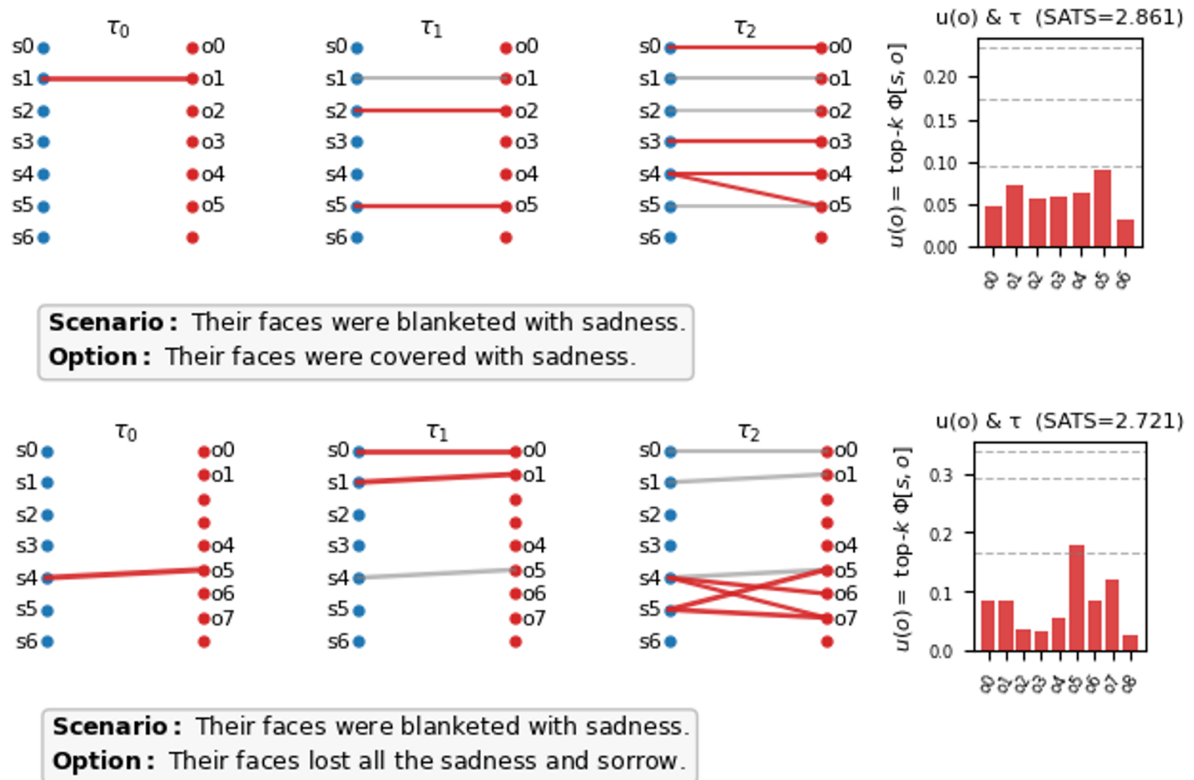


Figure 8: An error case of SATS on a FLUTE metaphor instance, illustrating multi-hop scenario–option interactions and option-level scores.